

AASHTO Census Transportation Solutions Technical Services Program

Census Transportation Planning Products: 2017 – 2021 Data

Key Considerations for Users

General Overview of CTPP Data

The Census Transportation Planning Products (CTPP) is a cooperative program funded by State Departments of Transportation that produces a set of special, transportation-focused tabulations using large sample surveys conducted by the U.S. Census Bureau. From 1970 to 2000, CTPP and its predecessor, the Urban Transportation Planning Package (UTPP), used data from the decennial Census long form. The decennial census long form was then replaced with a yearly survey called the American Community Survey (ACS), which provides additional transportation-related data that was once part of the Census long form. Currently, the data is released in 1-and 5-year datasets. CTPP utilizes the 5-year data to generate the primary datasets.

The 2006-2008 dataset was the first CTPP release to use the. It drew from the 3-year ACS data and was restricted to geographic units with a population of 20,000 or greater (counties and places). The 2006-2010 CTPP used 5-year ACS data and includes smaller geographic units, such as census tracts and traffic analysis zones (TAZs). The most recent CTPP releases are the 2017-2021 and 2012-2016 data sets, which are available on the American Association of State Highway and Transportation Officials (AASHTO) website: <https://ctpp.transportation.org/>.

The CTPP tabulations include three geographies:

- Part 1: Residence-based tables. These tables summarize the characteristics of people where they live.
- Part 2: Workplace-based tables. These tables summarize worker characteristics at their place of work.
- Part 3: Worker flows tables. Data on worker Origin to Destination flows between home and work, including travel mode.

Summary of Data Changes in 2017-2021

CTPP 2017-2021 data	CTPP 2012-2016 data
Data collected between 2017 and 2021.	Data collected between 2012 and 2016.
Dataset does not include data at the Traffic Analysis Zone (TAZ) or Transportation Analysis District (TAD) level. Tracts are the smallest level of geography available.	Includes data at the TAZ or TAD level, as well as tracts.
All tables were perturbed.	About half of the tables were perturbed.
For tables with 100 cells or more, small geographic entities are not included unless a suitable collapse can be created with fewer than 100 cells.	These rules were not in effect for CTPP 2012-2016.

Mean, Median, and Aggregate tables must have a minimum of 3 observations to be tabulated.	These rules were not in effect for CTPP 2012-2016.
ACS unweighted counts were protected starting in the ACS 2019 and 2015-2019 5-Year, and moving forward, using differential privacy by adding noise generated from a symmetric exponential distribution.	These rules were not in effect for CTPP 2012-2016.
The CTPP variable “Age of youngest person in the household” is no longer allowed, due to data disclosure concerns.	“Age of youngest person in the household” is available for CTPP 2012-2016.

Table Changes in 2017-2021

CTPP 2017-2021 data	CTPP 2012-2016 data
CTPP 2017-2021 Residence (Part 1) includes a new available topic: Population in Households	CTPP 2012-2016 does not include this topic.
CTPP 2017-2021 Workplace (Part 2) includes housing as an available topic.	CTPP 2012-2016 does not include this topic under Part 2.
CTPP 2017-2021 Workplace (Part 2) no longer includes PUMA as an available geography.	CTPP 2012-2016 includes PUMA as an available geography.
CTPP 2017-2021 Flows (Part 3) includes the topics of Housing and Populations and People.	CTPP 2012-2016 does not include these topics under Part 3.
CTPP 2017-2021 Flows (Part 3) includes the Tract -> State-County and State-County -> Tract geographies.	These geographies were not available for CTPP 2012-2016.

Geographic Levels

A key dimension of the CTPP is its geography. CTPP tables can be found for a variety of geographic areas, but not all tables are available for all areas.

Available Geography	Part 1	Part 2
Nation (US Total) (does not include PR)	X	X
States	X	
State (including Mexico, Canada, and Other)		X
State-County	X	X
State-County-MCD (12 MCD strong states)	X	X
State-Place	X	X
State-PUMA5	X	
Metropolitan Statistical Area-Each Principal City	X	X
State-County-Tract and Std ACS components	X	X
Block Groups for 5 published tables	X	

Flow Pairs (Part 3)

State	<i>to</i>	State
County	<i>to</i>	County
MCD	<i>to</i>	MCD
Place	<i>to</i>	Place
MSA	<i>to</i>	MSA
Place	<i>to</i>	County
County	<i>to</i>	Place
MCD	<i>to</i>	Place
PUMA5	<i>to</i>	Place
Tract	<i>to</i>	Tract
Tract	<i>to</i>	County
County	<i>to</i>	Tract

Universes

The current data set contains 20 distinct Universes. A universe is the complete set of individuals or entities that are being counted or studied. For CTPP for example, there are universes that distinguish between the total population and the population in just households. Of the 20 universes about half are for just workers, some being in Group Quarters and some not.

Disclosure and How It Impacts the Data

Before publishing any statistic, the U.S. Census Bureau applies safeguards to prevent someone from being able to trace that statistic back to a specific respondent. These safeguards are approved by the Census's Disclosure Review Board (DRB). The DRB required two disclosure processes on the CTPP data package to ensure that confidentiality and privacy is adequately protected. The first was applied by the Census Bureau using the rules outlined below.

- The following disclosure rules were applied to CTPP data by the Census Bureau:
 - o The results in all the tables are rounded.
 - o Perturbation techniques must be employed to account for unexpected deviations in results (see below for further discussion of perturbation techniques).
 - o Flows (part 3) must be subject to an unweighted minimum of 3 observations. This means 3 observations had to respond to the ACS that they live in one geography and work in the other.
 - o Non-residence geographies must have 50 unweighted cases to avoid suppression. This means that part 2 (workplace) data at small geographies will be in jeopardy of being suppressed.
 - o Mean, Median, and Aggregate tables must have a minimum of 3 observations to be tabulated

- Because of these requirements, the resulting data set is different in the following ways:
 - o Data is not available at the Transportation Analysis Zone (TAZ) or Transportation Analysis District (TAD) level for any variable.
 - o Data is not available at the block group level, except for a few variables
 - o When using the standard ACS microdata if a table is not available for a requested geography, it will be run using the perturbed microdata.
 - o If a table category is custom to CTPP and is not available or not a combination of a standard ACS published table category, the table will be subject to the Rule of 5 observations using standard ACS Microdata or Rule of 3 observations using perturbed microdata.

Perturbation & Suppression

Perturbation was the second process mandated by the DRB. Perturbation is a technique that is performed on microdata records prior to producing a frequency of cross-tabulating variables. Specifically, perturbation swaps values in one or more locations for variables on the microdata records. This introduces noise and uncertainty into the resulting tabulation that masks the identity of ACS respondents. To limit the “noise” there are bounds on the geographic areas in which the swapping occurs. Additionally, the weights are readjusted to preserve the integrity of the resulting weighted counts.

The entire package underwent perturbations, and many tables were subject to suppression. Suppression affected perturbed tables where there are three or fewer unweighted observations for a particular geographic entity in ACS.

With support from the National Highway Cooperative Research Program (NHCRP), and the American Association of State Highway and Transportation Officials (AASHTO), Westat developed and tested several disclosure avoidance techniques, including those that involve table-level modifications to already-generated tables, and those that focus on modifications to underlying microdata from which tables are derived. A technique was chosen and applied to the data. This methodology can be found https://ctppdata.transportation.org/CTPP_Methodology_Report.pdf.

Using the Data – Tips on using the software

- The most efficient way to find a data table of interest is first to use filters to narrow down to a set of tables and then use keyword search to find the tables of interest within that set.
- An Application Programming Interface (API) is available online here: <https://ctppdata.transportation.org/api-explorer.html>
 - o Questions? Send email to: ctpp-support@macrosysrt.com.

- While 2017-2021 CTPP data are available at a range of different geographies, keep in mind that the larger the geography, the larger the sample, and the lower the Margin of Error (MOE).
- Select data summaries might be available at small geographies in your area of interest but pay attention to the resulting MOEs and take note that perturbation is more prevalent when smaller geographies are being analyzed.

Additional Resources/Frequently Asked Questions

Q1. How do I access the CTPP 2017-2021 data?

A. The CTPP 2017-2021 data are available through our full data website:

<https://ctppdata.transportation.org/>

Q2. Is registration required to access the CTPP 2017-2021 data?

A. No. You can access the CTPP 2017-2021 data without registration, but to save your own geography selections, upload custom geographies for selections, or retrieve past work you MUST register. Registration is free for all users.

Q3. How does the CTPP 2017-2021 data product differ from the ACS Standard data product?

A. Key differences are:

- CTPP data products include journey to workflow data at large or small geographies while the ACS standard data product does not provide flow data.
- CTPP data products have workplace tables available at large or small geographies including Census Tract, but the workplace tables in standard ACS only go to the county level.
- The CTPP has many more transportation specific tables. CTPP 2017-2021 has 12 tables with commute mode in a cross-tabulation, while ACS standard products have only five tables.

Q4. How does the CTPP 2017-2021 data product differ from the CTPP 2012-2016 data product?

A. *See above tables (Summary of Data Changes in 2017-2021, Table Naming Changes in 2017-2021).*

Q5. How do the Census Bureau Disclosure Rules impact the CTPP 2017-2021 data?

A. Key differences are:

- Perturbation is required to produce the CTPP tables similarly to previous CTPP tabulations. *Perturbation is a technique that is performed on microdata records prior to producing a frequency of cross-tabulating variables. Perturbation swaps values in one or more locations for variables on the microdata records.* If a table is not available for a requested geography using standard ACS microdata, CTPP estimates use perturbed microdata.

- If a table category is custom to CTPP and not available or not a combination of a standard ACS published table category, the table is subject to the Rule of 5 observations using standard ACS Microdata or Rule of 3 observations using perturbed microdata.
- For tables with 100 cells or more, small geographic entities are not included unless a suitable collapse can be created with fewer than 100 cells. *For tables with 100-125 cells, only places with 10,000 or more population and higher-level geographies are included. For tables with more than 125 cells, only geographies with 100,000 or more population are included.*
- Mean, Median, and Aggregate tables must have a minimum of 3 observations to be tabulated.
- ACS unweighted counts (only produced for the Nation, State, County and Place for the 1-Year and 5-Year) were protected starting in the ACS 2019 and 2015-2019 5-Year, and moving forward, using differential privacy by adding noise generated from a symmetric exponential distribution.
- The CTPP variable “Age of youngest person in the household” is no longer allowed, due to data disclosure concerns.

Q6. Where can I find out more information about the quality of the data?

- A. The 2012-2016 CTPP data is based on the 2012-2016 American Community Survey products, which are the estimates of the entire population. All estimates produced from sample survey have sampling errors associated, the 2012-2016 CTPP product provides the Margin of Error for its estimates as measurements of the sampling error. To find out more about the quality of the data from the ACS Data Handbook: <https://www.census.gov/programs-surveys/acs/library/handbooks/general.html>

Q7. If the Margin of Error (MOE) is high, why should one bother using the results?

- A. If there is another data source that can provide reliable data for the geographic area, you may want to consider using that other source. However, if it is important to compare the results of different geographic areas to each other, one might still be better off using the ACS derived dataset since the survey methods have been applied consistently across all areas of the United States. Be sure to test the significance of any comparisons to ensure that they are statistically acceptable for your particular needs. *It is important to make sure that the margin of error (MOE) is not larger than the midpoint or value of the original data. For example, if the number of people who walk to work is 326 and the MOE is 350 you should not use the data.* For additional help, please visit the U.S. Census Bureau's ACS webpage: <https://www.census.gov/programs-surveys/acs/guidance.html>

Q8. Is there a resource on significance testing when comparing the CTPP 2017-2021 to 2012-2016, 2006-2010, 2000, 1990 or older Census results? Is it necessary to do any testing in these circumstances?

- A. The sample sizes for the 2000 decennial Census Long Forms were much larger than the ACS, even after 5 years of accumulating ACS responses. The associated MOEs for CTPP 2017-2021, 2012-2016 and 2006-2010 estimates are therefore larger in geographical areas containing relatively small sample sizes. “Looking for an easy way to conduct statistical testing? Try the Census Bureau's Statistical Testing Tool. Simply copy or download ACS estimates and their MOEs into the spreadsheet to get instant results of statistical tests:” <https://www.census.gov/programs-surveys/acs/guidance/statistical-testing-tool.html>

Supporting Documentation

[Technical Documentation \(census.gov\)](https://www.census.gov/programs-surveys/acs/technical-documentation.html) - <https://www.census.gov/programs-surveys/acs/technical-documentation.html>

[Statistical Testing Tool \(census.gov\)](https://www.census.gov/programs-surveys/acs/guidance/statistical-testing-tool.html) - <https://www.census.gov/programs-surveys/acs/guidance/statistical-testing-tool.html>

[Sample Size and Data Quality | American Community Survey | U.S. Census Bureau](https://www.census.gov/acs/www/methodology/sample-size-and-data-quality/) - <https://www.census.gov/acs/www/methodology/sample-size-and-data-quality/>

[ACS Resource Hub \(census.gov\)](https://www.census.gov/programs-surveys/acs/library/flyers/resource-hub.html) - <https://www.census.gov/programs-surveys/acs/library/flyers/resource-hub.html>

[2017-2021 PUMS 5-year Accuracy of the Data](https://usa.ipums.org/usa/resources/codebooks/AccuracyPUMS_1721.pdf) - https://usa.ipums.org/usa/resources/codebooks/AccuracyPUMS_1721.pdf

[2017-2021 American Community Survey-](https://storymaps.arcgis.com/stories/d0746b49c8ec4970b4e9fca0cf0f6aee) <https://storymaps.arcgis.com/stories/d0746b49c8ec4970b4e9fca0cf0f6aee>

[American Community Survey—Esri Demographics | Documentation-](https://doc.arcgis.com/en/esri-demographics/latest/esri-demographics/acs.htm) <https://doc.arcgis.com/en/esri-demographics/latest/esri-demographics/acs.htm>